7.5 Probing the Heart of Matter (PRO)

An area of fundamental physics that is the subject of current research, involving the acceleration and detection of high-energy particles and the interpretation of experiments:

- alpha scattering and the nuclear model of the atom
- accelerating particles to high energies
- detecting and interpreting interactions between particles
- the quark-lepton model.

Students will study the development of the nuclear model and the quark-lepton model to describe the behaviour of matter on a subatomic scale.

There are opportunities for students to develop ICT skills using the internet and computer simulations.

Students will learn how modern particle physics research is organised and funded, and hence have opportunities to consider ethical and other issues relating to its operation.

Students will be assessed on their ability to:		Suggested experiments
76	derive and use the expression $E_k = p^2/2m$ for the kinetic energy of a non-relativistic particle	
77	analyse and interpret data to calculate the momentum of (non-relativistic) particles and apply the principle of conservation of linear momentum to problems in one and two dimensions	
79	express angular displacement in radians and in degrees, and convert between those units	
80	explain the concept of angular velocity, and recognise and use the relationships $v = \omega r$ and $T = 2\pi/\omega$	
81	explain that a resultant force (centripetal force) is required to produce and maintain circular motion	
82	use the expression for centripetal force $F = ma = mv^2/r$ and hence derive and use the expressions for centripetal acceleration $a = v^2/r$ and $a = r\omega^2$	Investigate the effect of m, v and r of orbit on centripetal force

Students will be assessed on their ability to:		Suggested experiments
85	use the expression $F = kQ_1Q_2/r^2$, where $k = 1/4\pi\varepsilon_0$ and derive and use the expression $E = kQ/r^2$ for the electric field due to a point charge	Use electronic balance to measure the force between two charges
99	explain the role of electric and magnetic fields in particle accelerators (linac and cyclotron) and detectors (general principles of ionisation and deflection only)	
100	recognise and use the expression $r = p/BQ$ for a charged particle in a magnetic field	
101	recall and use the fact that charge, energy and momentum are always conserved in interactions between particles and hence interpret records of particle tracks	
102	explain why high energies are required to break particles into their constituents and to see fine structure	
103	recognise and use the expression $\Delta E = c^2\Delta m$ in situations involving the creation and annihilation of matter and antimatter particles	
104	use the non-SI units MeV and GeV (energy) and MeV/c^2, GeV/c^2 (mass) and atomic mass unit u, and convert between these and SI units	
105	be aware of relativistic effects and that these need to be taken into account at speeds near to that of light (use of relativistic equations not required)	
96	use the terms *nucleon number (mass number)* and *proton number (atomic number)*	
97	describe how large-angle alpha particle scattering gives evidence for a nuclear atom	
107	write and interpret equations using standard nuclear notation and standard particle symbols (e.g. π^+, e^-)	
106	recall that in the standard quark-lepton model each particle has a corresponding antiparticle, that baryons (e.g. neutrons and protons) are made from three quarks, and mesons (e.g. pions) from a quark and an antiquark, and that the symmetry of the model predicted the top and bottom quark	
108	use de Broglie's wave equation $\lambda = h/p$	